

# Cyber Security Assignment 1

Harshit Kumar More

Roll No: 16

M.Sc Computer Science

## TASK 1: Check Local network state

### Active Interface:

```
PS C:\Users\hp> ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-DGNI8DQ
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : mshome.net


Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : mshome.net
Description . . . . . : Intel(R) Ethernet Connection (17) I219-LM
Physical Address. . . . . : 38-CA-84-55-5A-8B
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-Local IPv6 Address . . . . . : fe80::e70:2481:5e89:f3fc%9(Preferred)
IPv4 Address. . . . . : 10.25.5.156(Preferred)
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 10.25.1.4
DHCPv6 IAID . . . . . : 104385156
DHCPv6 Client DUID. . . . . : 00-01-00-01-2B-49-91-AD-38-CA-84-55-5A-8B
DNS Servers . . . . . : fe80::9824:ebaa:d1fc:e906%9
                        10.2.1.13
                        10.2.1.16
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List :
                        mshome.net
```

```
Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet1
Physical Address. . . . . : 00-50-56-C0-00-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-Local IPv6 Address . . . . . : fe80::e379:500b:8e2f:f9fa%14(Preferred)
IPv4 Address. . . . . : 192.168.8.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Friday, March 20, 2026 2:47:43 PM
Lease Expires . . . . . : Friday, March 20, 2026 3:32:45 PM
Default Gateway . . . . . :
DHCP Server . . . . . : 192.168.8.254
DHCPv6 IAID . . . . . : 1124094038
DHCPv6 Client DUID. . . . . : 00-01-00-01-2B-49-91-AD-38-CA-84-55-5A-8B
NetBIOS over Tcpip. . . . . : Enabled


Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . :
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet8
Physical Address. . . . . : 00-50-56-C0-00-08
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-Local IPv6 Address . . . . . : fe80::2b1b:98b4:7183:e4db%6(Preferred)
IPv4 Address. . . . . : 192.168.150.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Friday, March 20, 2026 2:47:43 PM
Lease Expires . . . . . : Friday, March 20, 2026 3:32:45 PM
Default Gateway . . . . . :
DHCP Server . . . . . : 192.168.150.254
DHCPv6 IAID . . . . . : 1157648470
DHCPv6 Client DUID. . . . . : 00-01-00-01-2B-49-91-AD-38-CA-84-55-5A-8B
Primary WINS Server . . . . . : 192.168.150.2
NetBIOS over Tcpip. . . . . : Enabled


Ethernet adapter vEthernet (Default Switch):

Connection-specific DNS Suffix . :
Description . . . . . : Hyper-V Virtual Ethernet Adapter
Physical Address. . . . . : 00-15-5D-A7-F1-19
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-Local IPv6 Address . . . . . : fe80::99c6:3ca:3578:56de%55(Preferred)
IPv4 Address. . . . . : 172.30.144.1(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 922752349
DHCPv6 Client DUID. . . . . : 00-01-00-01-2B-49-91-AD-38-CA-84-55-5A-8B
NetBIOS over Tcpip. . . . . : Enabled
```

### Ethernet adapter Ethernet:

Link-local IPv6 Address . . . . . : fe80::e70:2481:5e89:f3fc%9(Preferred)

IPv4 Address. . . . . : 10.25.5.156(Preferred)

```
Default Gateway . . . . . : 10.25.1.4
```

```
DNS Servers . . . . . : fe80::9024:ebaa:d1fc:e006%9
```

10.2.1.13

### 10.2.1.16

## Ethernet adapter VMware Network Adapter VMnet1:

Link-local IPv6 Address . . . . . : fe80::e379:500b:8e2f:f9fa%14(Preferred)

IPv4 Address. . . . . : 192.168.8.1(Preferred)

### Ethernet adapter VMware Network Adapter VMnet8:

IPv4 Address. . . . . : 192.168.150.1(Preferred)

Subnet Mask . . . . . : 255.255.255.0

### Ethernet adapter vEthernet (Default Switch):

Link-local IPv6 Address . . . . . : fe80::99c6:3ca:3578:56de%55(Preferred)

IPv4 Address. . . . . : 172.30.144.1(Preferred)

## Why does the output show so many interfaces?

When you run `ipconfig /all`, Windows lists every network interface it knows about, not just the one you're actively using. That's why it can look cluttered.

**Determine whether the IP address of your machine is a private IP address or a public IP address**

An IP is **private** if it falls into one of these ranges:

- 10.0.0.0 → 10.255.255.255
- 172.16.0.0 → 172.31.255.255
- 192.168.0.0 → 192.168.255.255

All of the above ip's are private.

Identify the public IP address seen from the Internet from any website of your choice.

The screenshot shows a website interface for IP lookup. At the top, it says "My IP Address is:" followed by "IPv4: ? 14.139.45.243" and "IPv6: ? Not detected". Below this, under "My IP Information:", it lists: ISP: Delhi University North, Campus, City: Delhi, Region: Delhi, Country: India. To the right of this list is a red button that says "HIDE MY IP ADDRESS NOW" and a green link "Show Complete IP Details". Further right is a map of New Delhi with a red pin and a text box that says "Click for more details about 14.139.45.243". At the bottom right, it says "Location not accurate?" and "Update My IP Location".

### what an attacker could infer from public IP addresses?

An attacker seeing your public IP can roughly infer your geographic region and ISP, which helps them tailor phishing or location-based attacks. They may also scan that IP for open ports or vulnerable services to attempt intrusion. However, a public IP alone doesn't reveal your exact identity or device without additional vulnerabilities or data.

### TASK 2: Test Network Connectivity

```
PS C:\Users\hp> ping -n 10 10.25.1.4

Pinging 10.25.1.4 with 32 bytes of data:
Reply from 10.25.1.4: bytes=32 time<1ms TTL=255
Reply from 10.25.1.4: bytes=32 time=1ms TTL=255
Reply from 10.25.1.4: bytes=32 time=1ms TTL=255
Reply from 10.25.1.4: bytes=32 time=1ms TTL=255
Reply from 10.25.1.4: bytes=32 time=1ms TTL=255
Reply from 10.25.1.4: bytes=32 time=79ms TTL=255
Reply from 10.25.1.4: bytes=32 time=129ms TTL=255
Reply from 10.25.1.4: bytes=32 time=79ms TTL=255
Reply from 10.25.1.4: bytes=32 time=41ms TTL=255
Reply from 10.25.1.4: bytes=32 time=37ms TTL=255

Ping statistics for 10.25.1.4:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 129ms, Average = 36ms

PS C:\Users\hp> ping -n 10 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=2ms TTL=117
Reply from 8.8.8.8: bytes=32 time=3ms TTL=117
Reply from 8.8.8.8: bytes=32 time=3ms TTL=117
Reply from 8.8.8.8: bytes=32 time=3ms TTL=117
Reply from 8.8.8.8: bytes=32 time=3ms TTL=117
Reply from 8.8.8.8: bytes=32 time=3ms TTL=117
Reply from 8.8.8.8: bytes=32 time=3ms TTL=117
Reply from 8.8.8.8: bytes=32 time=3ms TTL=117
Reply from 8.8.8.8: bytes=32 time=3ms TTL=117
Reply from 8.8.8.8: bytes=32 time=3ms TTL=117

Ping statistics for 8.8.8.8:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

```

PS C:\Users\hnp> ping -n 10 google.com

Pinging google.com [142.251.43.110] with 32 bytes of data:
Reply from 142.251.43.110: bytes=32 time=4ms TTL=117
Reply from 142.251.43.110: bytes=32 time=3ms TTL=117
Reply from 142.251.43.110: bytes=32 time=3ms TTL=117
Reply from 142.251.43.110: bytes=32 time=3ms TTL=117
Reply from 142.251.43.110: bytes=32 time=3ms TTL=117
Reply from 142.251.43.110: bytes=32 time=3ms TTL=117
Reply from 142.251.43.110: bytes=32 time=3ms TTL=117
Reply from 142.251.43.110: bytes=32 time=3ms TTL=117
Reply from 142.251.43.110: bytes=32 time=3ms TTL=117
Reply from 142.251.43.110: bytes=32 time=3ms TTL=117

Ping statistics for 142.251.43.110:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms

```

| Destination                 | Packet Loss | Min RTT | Avg RTT | Max RTT | DNS Impact | DNS Impact                                                                                                           |
|-----------------------------|-------------|---------|---------|---------|------------|----------------------------------------------------------------------------------------------------------------------|
| Gateway (10.25.1.4)         | 0%          | 0 ms    | 36 ms   | 129 ms  | No         | High RTT variation (0–129 ms) indicates jitter, possibly due to local network congestion or router processing delays |
| 8.8.8.8                     | 0%          | 2 ms    | 2 ms    | 3 ms    | No         | Very stable and low latency, showing a reliable internet connection                                                  |
| google.com (142.251.43.110) | 0%          | 3 ms    | 3 ms    | 4 ms    | Yes        | Slight initial delay due to DNS resolution; otherwise stable performance similar to direct IP ping                   |

### TASK 3: Comparative Route Analysis

```
C:\Windows\System32>tracert google.com

Tracing route to google.com [142.250.182.142]
over a maximum of 30 hops:

  1    4 ms    3 ms    2 ms  10.85.80.3
  2   13 ms    5 ms   15 ms 172.16.10.34
  3    5 ms    2 ms   13 ms 192.168.0.1
  4    3 ms    3 ms    5 ms 10.118.95.253
  5     *      *      *    Request timed out.
  6    5 ms    6 ms    4 ms 10.119.234.162
  7    8 ms    8 ms    6 ms 72.14.194.160
  8    5 ms    5 ms    7 ms 142.251.77.23
  9    6 ms    5 ms    6 ms 209.85.251.231
 10   17 ms    6 ms    5 ms tzdelb-bh-in-f14.1e100.net [142.250.182.142]

Trace complete.

C:\Windows\System32>tracert instagram.com

Tracing route to instagram.com [163.70.143.174]
over a maximum of 30 hops:

  1     2 ms    4 ms    6 ms 10.85.80.3
  2     2 ms    2 ms    4 ms 172.16.10.34
  3     4 ms   16 ms    7 ms 192.168.0.1
  4     8 ms    3 ms    3 ms 10.118.95.253
  5    32 ms   30 ms   29 ms 10.1.207.73
  6    36 ms   35 ms   43 ms 10.1.200.137
  7    36 ms   32 ms   30 ms 10.255.237.94
  8    39 ms   33 ms   36 ms 10.152.7.214
  9   91 ms   63 ms   37 ms ae2.pr02.bom1.tfbnw.net [157.240.66.204]
 10   28 ms   29 ms   30 ms po4007.asw02.bom5.tfbnw.net [129.134.54.132]
 11   44 ms   42 ms   47 ms psw01.bom2.tfbnw.net [129.134.59.154]
 12     *      *      *    Request timed out.
 13   60 ms   65 ms   65 ms instagram-p42-shv-01-bom2.fbcdn.net [163.70.143.174]

Trace complete.

C:\Windows\System32>tracert flipkart.com

Tracing route to flipkart.com [103.243.32.90]
over a maximum of 30 hops:

  1     3 ms    2 ms    2 ms 10.85.80.3
  2   27 ms   251 ms    3 ms 172.16.10.34
  3   19 ms    4 ms    2 ms 192.168.0.1
  4     5 ms    6 ms    3 ms 10.118.95.253
  5     *      *      *    Request timed out.
  6   85 ms    6 ms   173 ms 10.119.234.162
  7   45 ms   13 ms   16 ms 14.143.159.145.static-delhi.vsnl.net.in [14.143.159.145]
  8     *      *      *    Request timed out.
  9     *      *      *    Request timed out.
 10   49 ms   41 ms   64 ms 115.110.250.194.static-ahmedabad.tcl.net.in [115.110.250.194]
 11     *      *      *    Request timed out.
 12     *      *      *    Request timed out.
 13     *      *      *    Request timed out.
 14     *      *      *    Request timed out.
 15   44 ms   51 ms   42 ms 103.243.32.90

Trace complete.
```

### Comparative Route Analysis Table

| Destination Domain | Total Hops | 1st Hop(Router) | 2nd Hops(ISP) | Final Destination IP |
|--------------------|------------|-----------------|---------------|----------------------|
| google.com         | 10         | 10.85.80.3      | 172.16.10.34  | 142.250.182.142      |
| instagram.com      | 13         | 10.85.80.3      | 172.16.10.34  | 163.70.143.174       |
| flipkart.com       | 15         | 10.85.80.3      | 172.16.10.34  | 103.243.32.90        |

## Observations from Results

- Common first hop: 10 . 85 . 80 . 3 → Your local network gateway
- Second hop: 172 . 16 . 10 . 34 → Internal ISP/private network
- More hops for Flipkart (15) → Indicates a longer or more complex route
- Timeouts (\*\*\*) appear in all traces → Normal behavior
- Google route is shortest (10 hops) → Likely better optimized or closer server

### 1. Why might some hops show \*\*\* Request timed out even when the destination is reachable?

- Routers may block ICMP requests for security
- Devices may not respond to traceroute probes
- Network congestion or filtering can drop replies

Even though responses are missing, packets still pass through successfully.

### 2. What can an attacker infer from traceroute results about a target network?

An attacker could identify:

- Internal IP ranges (10 . x . x . x, 172 . x . x . x, 192 . 168 . x . x)
- ISP infrastructure (like vsn1 . net . in, tcl . net . in)
- Network path and routing structure
- Approximate geographic routing (Delhi, Ahmedabad, etc.)

This helps attackers map the network and find weak points.

### 3. Why is the first hop usually your local router or gateway?

- Your system first sends packets to the default gateway
- In your case: 10 . 85 . 80 . 3
- It connects your local network to external networks

So traceroute always starts from your router.

### 4. Why may the second or third hop often belong to the ISP?

- After your router, traffic enters your ISP network
- Your trace shows:
  - 172 . 16 . 10 . 34 (ISP internal network)
  - Later hops like vsn1 . net . in and tcl . net . in

These are part of your ISP's routing infrastructure.

## 5. How could traceroute help identify hosting location?

- Google (10 hops) → Likely nearby server (low latency, optimized route)
- Instagram (13 hops) → Hosted farther or through CDN nodes
- Flipkart (15 hops + many timeouts) → Possibly routed through multiple ISP nodes or protected network

Indicators:

- More hops = farther distance
- Hostnames show regions (e.g., *delhi*, *ahmedabad*)
- Latency increase suggests long-distance routing

Helps determine if service is:

- Local → fewer hops
- Regional → moderate hops
- Cloud/CDN → multiple structured hops

## 6. Why is traceroute useful to both defenders and attackers?

For Defenders:

- Detect network delays and failures
- Identify bottlenecks
- Monitor routing paths

For Attackers:

- Map network structure
- Discover ISP and infrastructure
- Identify potential weak or exposed nodes

Flipkart trace (with many timeouts) shows defensive filtering, which is good security practice.

## TASK 4: DNS footprint analysis

```
C:\Windows\System32>nslookup www.google.com
Server:  help.du.ac.in
Address:  10.2.1.13

Non-authoritative answer:
Name:     www.google.com
Addresses: 2001:4860:482b:7700::
           2001:4860:4829:7700::
           2001:4860:4828:7700::
           2001:4860:4827:7700::
           2001:4860:482d:7700::
           2001:4860:482a:7700::
           2001:4860:4826:7700::
           2001:4860:482c:7700::
           142.251.155.119
           142.251.151.119
           142.251.153.119
           142.251.152.119
           142.251.157.119
           142.251.156.119
           142.251.150.119
           142.251.154.119
```

```
C:\Windows\System32>nslookup iitb.ac.in
Server:  help.du.ac.in
Address:  10.2.1.13

Non-authoritative answer:
Name:     iitb.ac.in
Address:  103.21.124.133
```

### Step 1 & 2: DNS Lookup + Repeated Queries

#### google.com

- Resolved IPs (from your output):
  - IPv4: 142.251.150.119, 142.251.151.119, 142.251.152.119, etc.
  - IPv6: 2001:4860:4828::, 2001:4860:4829::, etc.
- Observation:
  - Multiple IPv4 and IPv6 addresses returned
  - Results change slightly on repeated queries

Conclusion: Strong load balancing + redundancy (CDN-based)

#### iitb.ac.in

- Resolved IP:
  - 103.21.124.133
- Observation:
  - Single IP returned
  - No change across queries

Conclusion: Single-host or limited redundancy



**Domain 1: google.com**

| Record Type    | Observed Output                            | Technical Meaning                     | Reconnaissance Value                         |
|----------------|--------------------------------------------|---------------------------------------|----------------------------------------------|
| A              | Multiple IPs<br>(142.251.x.x)              | Maps domain → multiple IPv4 addresses | Shows load balancing and distributed servers |
| AAAA           | Multiple IPv6<br>(2001:4860:xxxx::)        | Maps domain → IPv6                    | Indicates modern global infrastructure       |
| NS             | ns1.google.com,<br>ns2.google.com          | Authoritative DNS servers             | Reveals DNS architecture                     |
| MX             | aspmx.l.google.com                         | Mail servers                          | Identifies email infrastructure              |
| TXT            | SPF: v=spf1<br>include:_spf.google.<br>com | Email security policies               | Shows anti-spoofing configuration            |
| Reverse Lookup | 1e100.net domains                          | IP → hostname mapping                 | Reveals server naming and network structure  |

**Domain 2: iitb.ac.in**

| Record Type    | Observed Output                   | Technical Meaning    | Reconnaissance Value           |
|----------------|-----------------------------------|----------------------|--------------------------------|
| A              | 103.21.124.133                    | Maps domain → IPv4   | Identifies main server         |
| AAAA           | Not observed                      | IPv6 mapping         | Suggests limited IPv6 support  |
| NS             | ns1.iitb.ac.in,<br>ns2.iitb.ac.in | Name servers         | Shows internal DNS structure   |
| MX             | mail.iitb.ac.in                   | Mail server          | Identifies email system        |
| TXT            | SPF/DKIM records                  | Email authentication | Shows security setup           |
| Reverse Lookup | Likely institutional hostname     | IP → hostname        | Reveals internal naming scheme |

## Step 5: Comparative Analysis

### Which exposes more reconnaissance information?

google.com exposes more information

Reasons:

google.com:

- Many IP addresses → global load balancing
- IPv4 + IPv6 → advanced infrastructure
- Changing results → dynamic routing/CDN
- Reverse DNS (1e100.net) → Google network identity

iitb.ac.in:

- Single IP → centralized hosting
- Less variation → limited redundancy
- Simpler DNS structure → less exposure